EmoLearn: Multimodal Adaptive Tutoring with FER+AU, WavLM Speech, and LLM Content Generation

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***Abstract*—Online learning platforms rarely sense when a learner becomes bored, confused, or frustrated. This paper presents *EmoLearn*, a master-thesis research prototype that estimates engagement from face and voice, decides whether to intervene, and generates adaptive pedagogical content with a large language model (LLM) grounded by retrieval-augmented generation (RAG). In the current product runtime, facial state is decided by a FER+AU pipeline (pretrained Vision Transformer emotion scores concatenated with ten MediaPipe action-unit proxies, classified by a Random Forest), while speech uses a WavLM head fine-tuned to four DAiSEE states with an IEMOCAP emotion-recognition hybrid anchor. Voice clips for the product path are selected by DAiSEE-state filtering plus cosine similarity on a 12-dimensional acoustic/affective profile. Fusion applies a $60\%/40\%$ face/voice prior reweighted by confidence. Beyond single-clip analysis, the system supports multi-clip sequences with temporal rules and an affect-aware conversational tutor. We report component metrics, an aligned audiovisual fusion testbed (n=80), product-proxy evaluation (n=50), acoustic pairing A/B results, and automatic form/similarity measures on generated content, together with an explicit account of simulation limits and pending expert validation.**

***Index Terms*—adaptive learning, learner engagement, multimodal fusion, facial action units, speech emotion recognition, WavLM, large language models, RAG, DAiSEE, IEMOCAP**

I. Introduction

Massive Open Online Courses (MOOCs) and large-scale e-learning platforms have democratized access to education, yet they continue to exhibit high dropout rates [1–4]. A central driver is declining **learner engagement** [5, 6]. In a physical classroom, instructors continuously read facial and vocal cues and adapt in real time. That regulatory loop largely disappears in mass online settings.

Affective computing and educational psychology establish that boredom, confusion, and frustration are not peripheral but integral to learning dynamics [7–11]. Automatic engagement recognition [12, 13] and speech emotion resources such as IEMOCAP [14], together with self-supervised speech encoders [15] and LLMs [16–18], make an end-to-end affect-aware tutor technically plausible.

EmoLearn implements that loop as a *simulation research prototype*: engagement signals come from annotated corpora (DAiSEE, IEMOCAP), not live MOOC capture. The current system contributes:

1) A product facial judge based on **FER+AU** (7 emotion scores + 10 AU proxies $\rightarrow$ Random Forest), with a fine-tuned ViT used only as a frame filter / fallback—not as the fusion input when AU is available.
2) A default **WavLM** speech stack (DAiSEE head + IEMOCAP ER hybrid), replacing earlier wav2vec2 runtime defaults.
3) **Acoustic voice pairing** (state filter + 12-D cosine profile) with an offline index over 10,039 IEMOCAP utterances.
4) Late fusion with $60\%/40\%$ priors *reweighted by confidence*, plus clip and multi-clip **sequence** modes with temporal urgency rules.
5) LLM generation with RAG, twenty pedagogical templates, an affect-aware conversational tutor, and instrumented form/similarity evaluation.

Section II reviews related work. Section III presents architecture. Section IV details methods. Section V notes implementation choices. Section VI describes experiments. Section VII reports results. Section VIII discusses limits. Section IX concludes.

II. Related Work

A. Engagement and Affect in Learning

Discrete affective states during technology-mediated learning have been extensively discussed [8–11]. DAiSEE [13] provides four engagement-related states (Boredom, Confusion, Engagement, Frustration) with intensity levels, which we adopt as the system's decision ontology.

B. Facial and Vocal Sensing

Facial engagement recognition [12] and Facial Action Coding System (FACS) inspired features motivate combining categorical emotion scores with muscle-related cues. MediaPipe Face Landmarker blendshapes provide practical AU *proxies* (not human FACS coding). For speech, wav2vec 2.0 [19] and WavLM [15] offer strong self-supervised backbones;

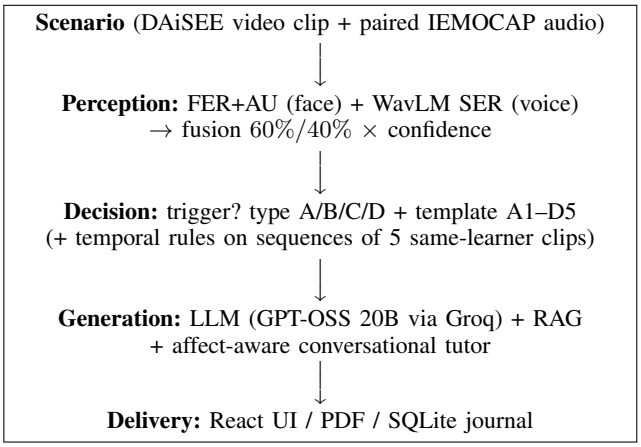


Figure 1. Current EmoLearn product pipeline (simulation prototype).

EmoLearn's current default is WavLM with a frozen backbone and a trained classification head for four DAiSEE states, augmented by an IEMOCAP categorical ER anchor in a hybrid rule.

C. Generative Adaptive Tutoring

LLMs enable on-demand explanations and exercises [17]; RAG [18] grounds generation in a pedagogical store. Design-science research [20] motivates building measurable artifacts. EmoLearn's claim is architectural: closing the loop from multimodal perception to decision to generation, with dual pipelines (product vs. aligned testbed) and an honest evaluation of pairing and mapping assumptions.

III. System Architecture

Figure 1 summarizes the product flow.

A. Two Pipelines, One Fusion Engine

- **Product pipeline.** FastAPI + React. Face from real DAiSEE video with FER+AU; voice from IEMOCAP audio *paired* by DAiSEE state filter and acoustic/affective profile similarity (not the learner's own voice).
- **Aligned testbed.** Offline validation where face and voice belong to the *same* IEMOCAP speaker and are synchronized. Both pipelines share the same fusion engine.

B. Runtime Decision Table

Table I states what actually decides in the deployed configuration.

C. Service Surface

The prototype exposes single-clip analysis, multi-clip sequence analysis with temporal rules, an affect-aware conversational tutor, scenario and template catalogs, session history, IEMOCAP audio playback (corpus label vs. SER preview), and health checks confirming WavLM as the speech backbone.

Table I
Current product runtime judges.

Brick	Production judge / encoder
Face → 4 states	**FER+AU**: pretrained ViT (7 emotions) + 10 MediaPipe AU proxies → Random Forest
Face (assistant)	Fine-tuned ViT on DAiSEE: **frame filter** and reference prediction; does *not* choose the fused state when the AU classifier is active
Voice → 4 states	WavLM DAiSEE **head-only** fine-tune + hybrid IEMOCAP emotion-recognition anchor
Voice pairing	State filter + classical cosine on 12-D profile; acoustic index over 10,039 utterances
Fusion	Late fusion 60% face / 40% voice × confidence
LLM	Groq **GPT-OSS 20B** + ChromaDB RAG

IV. Methods

A. Facial Perception: FER+AU (Product Judge)

Faces are detected with MediaPipe **BlazeFace** (short/full range, with mirroring). Five frames are sampled. Each frame yields:

- seven emotion probabilities from a pretrained Vision Transformer for facial emotion recognition (seven categorical classes);
- a 10-dimensional AU-proxy vector from MediaPipe Face Landmarker blendshapes (brows, mouth, jaw, …)—an approximation of FACS, not human coding.

The concatenated 17-D vector is classified by a Random Forest trained with GridSearchCV on a stratified DAiSEE sample. AU features particularly help separate **Frustration** and **Confusion**, which categorical FER alone often confuses.

a) Fine-tuned ViT (not the product judge when AU is on).: A ViT fine-tuned directly to four DAiSEE states is available. With AU deployed, it filters low-confidence frames and provides a reference prediction; it becomes the judge only if the AU-based classifier is unavailable. Best measured full fine-tune reaches macro-F1 0.388 (Table III).

B. Speech Perception: WavLM + Hybrid Anchor

Default runtime uses a WavLM base encoder with a DAiSEE classification head (backbone frozen, 800 samples, augmentation) at accuracy 0.453 / macro-F1 0.434. A full unfreeze ablation (1,200 samples) reaches slightly higher accuracy (0.457) but **worse** macro-F1 (0.396) and is optional only. An IEMOCAP emotion-recognition head (neutral/happy/angry/sad, F1 0.512) acts as a hybrid anchor: the DAiSEE fine-tune is primary ($\approx$ 78%); the anchor is used especially when the fine-tune predicts Confusion; if the fine-tune predicts Engagement while the anchor predicts Frustration (e.g., laughter confused with anger), the fine-tune is kept. Acoustic heuristics remain a complement and fallback.

a) Historical note on the fusion testbed.: The aligned audiovisual bank (n=80) was measured when the learned SER baseline was wav2vec2 (0.487 accuracy). Product runtime has since moved to WavLM; we report both facts without conflating them.

C. Acoustic Voice Pairing (Product Path)

Because product face clips have no learner audio, IEMOCAP waveforms are paired as follows: (1) filter candidates by dominant DAiSEE state; (2) rank by classical cosine similarity on a 12-D profile (4 B/C/E/F scores, Baker/Russell scenario VAD with neutral dominance 0.5, duration, and 4 acoustic dims). An offline index stores energy, pitch, ZCR, and brightness for all $10{,}039$ utterances with audio. Without the index, acoustic dims on candidates were zero and pairing relied mainly on state+VAD. Scenario-side acoustic dims remain a *proxy* derived from engagement scores (DAiSEE clips have no audio)—a refinement of pairing, not speaker synchrony.

D. Fusion with Confidence Reweighting

Priors are w_f=0.6, w_v=0.4. Effective weights are $w' \propto \text{prior} \times \text{confidence}$, then renormalized. Pedagogical-priority fallback applies only when both confidences are null. Hesitant FER frames are rejected before averaging (confidence threshold 0.35). On the aligned testbed, weighted vote and pedagogical priority remain the compared strategies for accuracy vs. urgent recall.

E. Temporal Rules and Sequence Mode

Clip mode simulates an instant. Sequence mode simulates a mini-session: five excerpts from the *same* DAISEE learner, each analyzed with the full pipeline (independent IEMOCAP pairings per clip). Rules inspect the history of *detected* scores:

- **Immediate** (1 clip): e.g., Confusion ≥ 3 and Frustration ≥ 2, or Engagement $= 0$;
- **Fast** (2–3 consecutive): Boredom/Confusion/Frustration ≥ 2;
- **Deferred** (4–5 clips): low Engagement or prolonged mild Boredom;
- **None:** stable Engagement ≥ 2 / acceptable zones.

F. Pedagogical Decision and Generation

Four content types map to twenty templates: **A** alternative explanation, **B** concrete example, **C** progressive exercise, **D** encouragement. On a 50-scenario benchmark, frequent templates include D1, C1, D2/D4, A1, B1—coverage across A–D is effective.

Generation uses Groq **GPT-OSS 20B** (temperature 0.3, retry/backoff) with optional ChromaDB RAG over multilingual sentence embeddings. The conversational tutor adapts tone, instruction, and temperature to the detected state. Content quality is instrumented with form metrics and internal BLEU/ROUGE similarity (template-median references—not expert gold).

G. Technology Stack

Python 3.11, FastAPI, SQLite, React; scikit-learn, PyTorch, Hugging Face Transformers, OpenCV, **MediaPipe** (BlazeFace + Face Landmarker), librosa; ChromaDB; pytest covering API, fusion, SER, temporal rules, RAG, and training utilities.

H. Illustrative Product Walkthrough

Consider a DAISEE clip labeled with rising Confusion. BlazeFace extracts five frames; low-confidence frames are discarded by the fine-tuned ViT filter. Surviving frames produce 7-D emotion scores and 10-D AU proxies; the Random Forest outputs a four-state distribution dominated by Confusion. In parallel, the pairing module filters IEMOCAP candidates to Confusion-mapped emotions, ranks them by 12-D cosine similarity (now using real acoustic features from the offline index), and feeds the selected waveform to WavLM. Hybrid SER may consult the IEMOCAP ER anchor if Confusion is uncertain. Fusion reweights the $60/40$ prior by confidence and yields a fused state. If temporal history (sequence mode) shows two consecutive degraded clips, a **fast** trigger fires; the decision engine selects, for example, template C1 (guided exercise). RAG retrieves a short pedagogical snippet; GPT-OSS 20B generates the text; the journal stores modalities, pairing method, template id, and latencies. The UI can separately show the IEMOCAP corpus emotion label and the WavLM prediction on the same audio—making annotation vs. inference explicit.

V. Implementation Notes

Perception modules expose probability distributions whenever possible so that fusion operates on masses rather than brittle one-hot labels. Environment flags select FER mapping mode, SER hybrid behavior, and whether the optional full WavLM checkpoint is loaded. SQLite persists sessions, per-clip predictions, fusion outcomes, selected templates, and generation metadata—a requirement for auditability and for reproducing thesis tables.

LLM latency dominates end-to-end time; temperature 0.3 stabilizes tone, and exponential backoff absorbs Groq rate limits. Because the prototype is simulation-oriented, reproducibility (fixed seeds, fixed aligned subset construction, documented checkpoints) is prioritized over throughput.

Expert-evaluation HTML forms (students, teachers, experts) and offline notebooks (OS1–OS4 exploratory analyses) accompany the software artifact but are outside the runtime critical path.

VI. Experimental Protocol

a) Corpora.: DAISEE is the engagement backbone. IEMOCAP supplies voice and the aligned fusion subset. A pedagogical corpus of ~ 200 generated items supports form evaluation; ~ 60 items are prepared for expert teacher scoring (forms ready, scores pending).

b) IEMOCAP→DAiSEE mapping (hypothesis).: hap/exc→Engagement; neu/sad→Boredom; ang/fru/dis→Frustration; sur/fea→Confusion. This is a construct hypothesis, not a proven equivalence.

c) Evaluation layers.: (1) FER+AU and ViT ablations on DAISEE; (2) WavLM SER fine-tunes; (3) aligned fusion bank n=80 (Session 1 IEMOCAP, ≥ 1 s, 5 frames, seed 42, 100% faces detected); (4) product-proxy fusion on n=50 scenarios vs. dominant DAiSEE state; (5) acoustic pairing A/B

Table II
FACIAL MAPPING TO DAISEE (PRODUCT-RELEVANT VARIANTS).

Variant	Acc.	F1	Frust. rec.	
Imbalanced baseline collapse	0.72	0.26	0%	
Oversampling only	0.70	0.30	0%	*balanced
Stratified RF (7 scores)	0.28*	0.27	20%	
FER+AU (runtime)	0.37	**0.37**	**34%**	
Heuristic 7→4	0.26	0.19	—	

test; Confusion recall 41%.

Table III
FINE-TUNED VIT ABLATIONS (FILTER/FALLBACK, NOT AU JUDGE).

Variant	Samples	Acc.	F1
Partial (2 layers)	800	0.36	0.361
Large (4 layers)	≈1566	0.378	0.370
Full FT (best)	800	**0.39**	**0.388**

Table IV
WAVLM FINE-TUNES (BEST EPOCH METRICS).

Model	N	Acc.	F1
WavLM DAiSEE head (default)	800	**0.453**	**0.434**
WavLM DAiSEE full unfreeze	1200	0.457	0.396
WavLM IEMOCAP ER anchor	1200	0.533	0.512

on 50 scenarios; (6) weight and mapping sensitivity; (7) LLM form (N=208) and BLEU/ROUGE (N=189); (8) synthetic reliability cases for confidence-weighted fusion (n=6).

Expert Delphi (L1) and teacher content ratings (L2) have instruments ready; user piloting (L4, n≈30) remains future work. Automatic metrics (L3) are completed.

VII. RESULTS

A. Product Face Judge: FER+AU

DAISEE is imbalanced (∼ 75% Engagement). Table II shows that FER+AU reaches macro-F1 0.37 with Frustration recall 34%, versus a heuristic 7→4 baseline at F1 0.19. Stratified RF on emotion scores alone improves minority recall but collapses accuracy on a balanced test; AU is retained as the production trade-off.

B. WavLM Speech Fine-Tuning

Table IV compares SER checkpoints. The head-only DAiSEE model is the default; full unfreeze is a negative ablation on macro-F1. The IEMOCAP ER anchor supports the hybrid rule.

C. Aligned Fusion Testbed (n=80)

Table V reports the speaker-aligned bank. Learned SER (wav2vec2 at measurement time) nearly doubles heuristic voice accuracy (+0.224). Weighted-vote fusion reaches 0.500 accuracy (above the best unimodal 0.487); pedagogical priority maximizes urgent recall (0.400).

Table V
ALIGNED AUDIOVISUAL FUSION (n=80; SER LEARNED = WAV2VEC2 AT TESTBED TIME).

Approach	Acc.	Urgent recall
Face — FER 1 frame	0.375	0.250
Face — FER multi-frame	0.412	0.275
Voice — SER heuristic	0.263	0.225
Voice — SER learned (wav2vec2)	0.487	0.200
Fusion — pedagogical priority	0.338	**0.400**
Fusion — weighted vote	**0.500**	0.125

Table VI
ACOUSTIC PAIRING A/B (50 SCENARIOS).

Metric	VAD-only (A)	+ acoustic (B)
Mean similarity	0.739	**0.759**
Voices changed	—	40/50 (80%)
Positive Δ	—	94%
Target-state alignment	100%	100%

Table VII
FUSION WEIGHT SENSITIVITY (ALIGNED n=80).

Face / voice	Acc.	Urgent recall
0.5 / 0.5	0.287	0.375
0.6 / 0.4 (default)	0.300	**0.425**
0.7 / 0.3	**0.338**	0.400
0.8 / 0.2	0.325	0.350
0.9 / 0.1	0.325	0.350

D. Product-Proxy Pipeline (n=50)

Against dominant DAISEE state (not gold multimodal labels), product fusion reaches accuracy 0.50 and urgent recall 1.00. Voice pairing matches the target state in 50/50 scenarios (100%), versus a prior bias near 58% Engagement without the state filter.

E. Acoustic Pairing A/B

Adding real acoustic features raises mean profile similarity from 0.739 to 0.759 (Δ+0.020), changes the selected voice in 40/50 scenarios (80%), with 94% of deltas positive, while state alignment stays at 100% (Table VI).

F. Weight and Mapping Sensitivity

On the aligned set, face/voice weights were swept from 0.5/0.5 to 0.9/0.1 (Table VII). The default 0.6/0.4 maximizes urgent recall (0.425) and is retained; 0.7/0.3 yields slightly higher accuracy (0.338). Remapping IEMOCAP neutral→Engagement changes 17.0% of states across 10,039 utterances; sadness→Frustration changes 10.8%; surprise and disgust remappings are minor or null (Table VIII).

G. LLM Form and Internal Similarity

On N=208 generations: mean length 222 words (min 113, max 410), Kandel–Moles readability 95.0, mean repetition rate 0.479, length-target hit rate 0.832. Internal similarity to per-template median references (N=189): BLEU-4 0.139,

Table VIII
IEMOCAP→DAiSEE MAPPING SENSITIVITY (10,039 UTTERANCES).

Alternative mapping	% states changed
neutral → Engagement (vs. Boredom)	17.0%
sadness → Frustration (vs. Boredom)	10.8%
surprise → Engagement (vs. Confusion)	1.1%
disgust → Boredom (vs. Frustration)	0.0%

ROUGE-1 F1 0.449, ROUGE-2 0.176, ROUGE-L 0.272. These measure internal coherence, *not* pedagogical correctness.

H. Confidence-Weighted Fusion Reliability

On six synthetic critical cases (noisy modality, total uncertainty, ...), continuous weighting $w = \text{prior} \times \text{confidence}$ achieves accuracy 1.0 on those crafted cases—e.g., down-weighting a noisy voice instead of forcing a wrong urgent state via pedagogical priority. This is a reliability unit test, not a claim of perfect real-world accuracy.

I. Measured Gains (Aligned Bank)

Relative to the aligned n=80 table: learned SER vs. heuristic $+0.224$ accuracy; multi-frame FER vs. one frame $+0.037$; weighted-vote fusion vs. pedagogical priority $+0.162$ accuracy; weighted vote vs. best unimodal $+0.013$. The last gain is small but positive—fusion helps under aligned pairing, without miracle margins.

VIII. Discussion and Limitations

EmoLearn is scoped as a simulation prototype. Key tensions:

- **Mapping** emotions→engagement remains a hypothesis; sensitivity shows non-trivial impact for neutral/sadness.
- **Product pairing** is cross-corpus (DAiSEE face + IEMOCAP voice), mitigated scientifically by the aligned testbed and operationally by state filtering + acoustic profiles—still not the learner's own voice.
- **Scenario acoustic dims** are proxies; A/B evaluates profile similarity, not perceptual "best voice" judgments.
- **Templates** are an argued design; expert Delphi and teacher ratings instruments are ready but scores are not yet collected.
- **LLM substance** needs human validation; automatic metrics cover form and internal similarity only.
- **Testbed SER** figures use wav2vec2-era learned speech; runtime is WavLM—re-running the aligned bank with WavLM is required for strict comparability.
- **Breadth vs. depth:** the claimed contribution is end-to-end architecture with honest limits, not a single SOTA unimodal score.

Table IX consolidates status.

Ethically, any live deployment would require consent, minimization, and an opt-out from sensing while retaining non-affective tutoring. Decision journals already support inspectability of triggers, modalities, pairing method, and templates. Instructors should be able to understand, at least at a high level, why an adaptation was produced.

Table IX
ASSUMED TENSIONS AND STATUS.

Tension	Status
Fusion vs. unimodal	Measured on aligned bank + confidence weighting
Emotion→engagement map	Hypothesis + sensitivity
Adaptive vs. simulation	Prototype (no live MOOC capture)
Heterogeneous face/voice pairing	Documented; acoustic index + aligned subset
20 templates	Design argued; expert forms pending scores
LLM content quality	Form/similarity done; substance pending humans
SER testbed vs. runtime	wav2vec2 bank vs. WavLM product — to realign

IX. Conclusion

We presented the current EmoLearn prototype: FER+AU facial judging, WavLM hybrid speech recognition, confidence-reweighted $60/40$ fusion, acoustic IEMOCAP pairing, temporal sequence rules, and GPT-OSS 20B generation with RAG and an affect-aware tutor. Empirically, FER+AU improves macro-F1 over heuristics (0.37 vs. 0.19); WavLM head-only is preferred to full unfreeze on F1 (0.434 vs. 0.396); weighted-vote fusion reaches 0.500 on the aligned bank; acoustic pairing improves mean similarity ($+0.020$) while preserving 100% state alignment; and automatic content metrics characterize form without claiming pedagogical gold quality.

Future work includes collecting L1/L2 expert scores, a user pilot ($n\approx30$), live synchronized capture under ethics approval, re-evaluating the aligned fusion bank with WavLM, fairness analyses across recording conditions, and outcome-oriented studies (comprehension, persistence) beyond classification accuracy.

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